

LAB2 Report

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In lab2 we explored the design of ECG signal processing and had a nice journey exploring the oversampling techniques.

1 Hardware configuration

2 Amplifier Frequency Response

3 Signal Processing

3.1 Direct Digital Filtering on Oversampled Signal

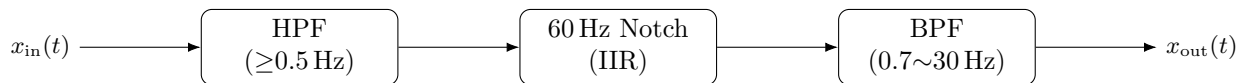


Figure 1: Direct digital processing with the oversampled signal

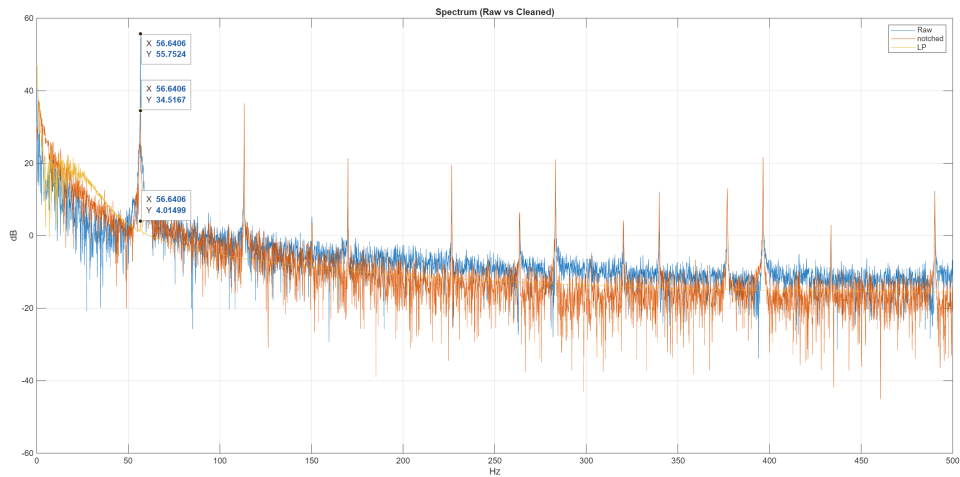


Figure 2

Listing 1: Baseline and 60 Hz cleanup with bandpass filtering.

```
1 %% ===== 2) Baseline & 60 Hz cleanup =====
2 hpFc = 0.7; % ECG above ~0.7 Hz
3 hp = designfilt('highpassiir','FilterOrder',4, ...
4     'HalfPowerFrequency',hpFc,'SampleRate',Fs,'DesignMethod','butter');
5
6 notch60 = designfilt('bandstopiir','FilterOrder',2, ...
```

```

7      'HalfPowerFrequency1',56,'HalfPowerFrequency2',57, ...
8      'DesignMethod','butter','SampleRate',Fs);
9
10 x_hp = filtfilt(hp, x);
11 x_n60 = filtfilt(notch60, x_hp);
12
13 %% ===== 3) Bandpass 0.7~30 Hz =====
14 bp = designfilt('bandpassiir','FilterOrder',6, ...
15     'HalfPowerFrequency1',0.7,'HalfPowerFrequency2',30, ...
16     'DesignMethod','butter','SampleRate',Fs);
17 x_bp = filtfilt(bp, x_n60);

```

3.2 Downsampling

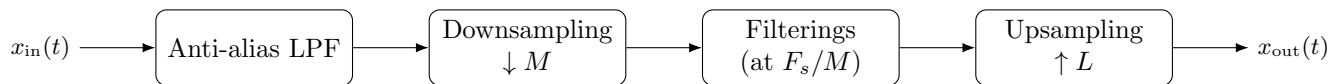


Figure 3: Signal Processing Pipeline with Downsampling

Listing 2: Anti-alias low-pass filtering before downsampling.

```

1 aa = designfilt('lowpassiir','FilterOrder',8, ...
2     'HalfPowerFrequency', 0.9*(Fs_ds/2), ...
3     'DesignMethod','butter','SampleRate',Fs);
4 x_aa = filtfilt(aa, x);

```

We can see, as in our digital signal processing lectures suggested, the downsampling in time domain gives us a stretch in the frequency domain.

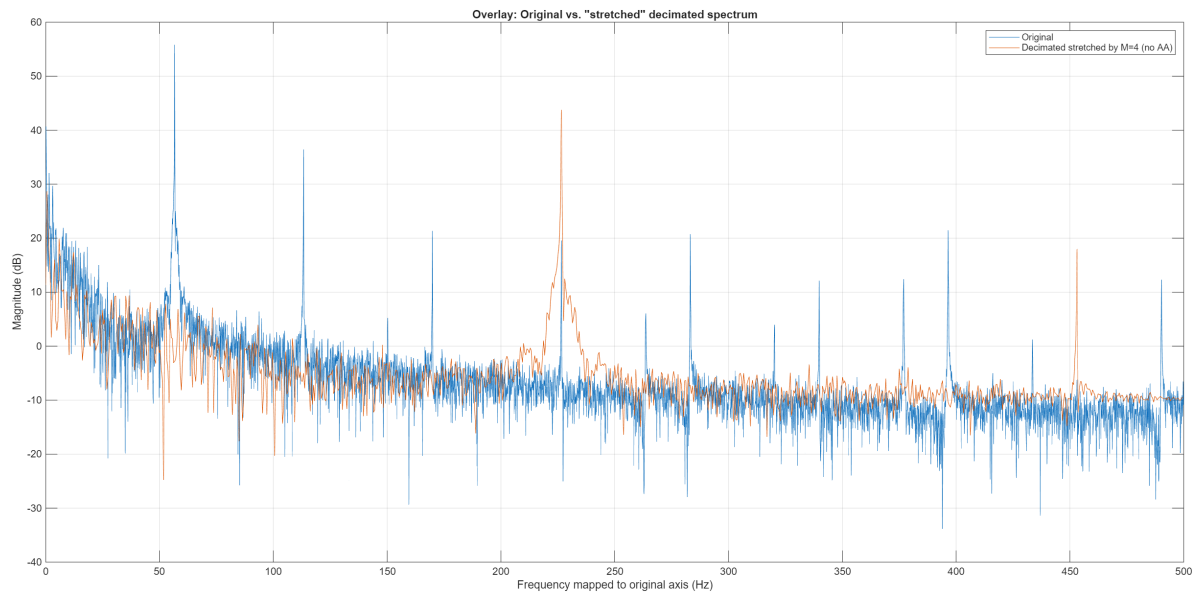


Figure 4

Listing 3: Direct and averaged downsampling.

```

1 % Direct downsampling
2 x_ds = x_aa(1:M:end);
3 t_ds = t(1:M:end);
4
5 % Averaged downsampling
6 Lfull = floor(numel(x)/M)*M;
7 x_trunc = x_aa(1:Lfull);
8 x_blocks = reshape(x_trunc, M, []);

```

```
9 x_ds_avg = mean(x_blocks,1).';
```

We can see an obvious difference on the ambient noise level. (with the normalized frequency axis by F_s/M)

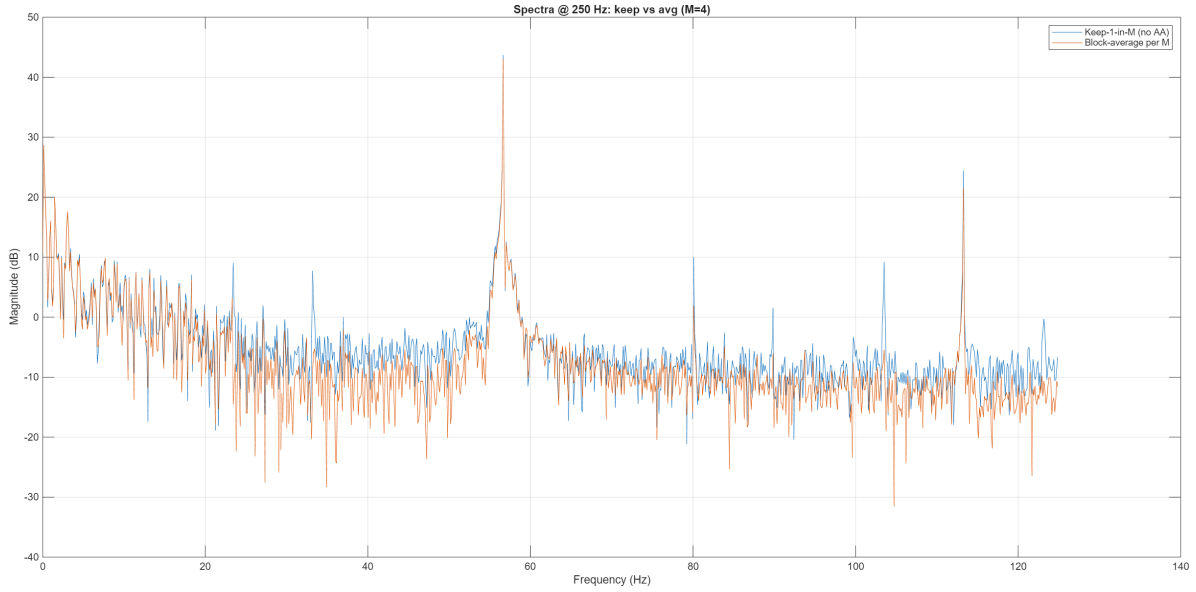


Figure 5

We then apply the same filtering design on the downsampled signal.

Listing 4: High-pass, notch, and band-pass filtering at the reduced sampling rate.

```
1 hp_ds = designfilt('highpassiir','FilterOrder',4, ...
2   'HalfPowerFrequency',0.7,'SampleRate',Fs_ds,'DesignMethod','butter');
3 bp_ds = designfilt('bandpassiir','FilterOrder',6, ...
4   'HalfPowerFrequency1',0.7,'HalfPowerFrequency2',30, ...
5   'DesignMethod','butter','SampleRate',Fs_ds);
6 notch_ds = designfilt('bandstopiir','FilterOrder',2, ...
7   'HalfPowerFrequency1',56,'HalfPowerFrequency2',57, ...
8   'DesignMethod','butter','SampleRate',Fs_ds);
9
10 x_ds_hp = filtfilt(hp_ds, x_ds);
11 x_ds_n60 = filtfilt(notch_ds, x_ds_hp);
12 x_ds_bp = filtfilt(bp_ds, x_ds_n60);
13
14 x_ds_hp_avg = filtfilt(hp_ds, x_ds_avg);
15 x_ds_n60_avg = filtfilt(notch_ds, x_ds_hp_avg);
16 x_ds_bp_avg = filtfilt(bp_ds, x_ds_n60_avg);
```

Listing 5: Reconstructing the signal to 1 kHz via interpolation.

```
1 x_up = interp1(t_ds, x_ds_bp, t_trim, 'pchip', 'extrap');
2 x_up_avg = interp1(t_ds, x_ds_bp_avg, t_trim, 'pchip', 'extrap');
```

We can observe that the downsampling signals kept more details of the ECG signals, especially obvious on the wiggles of the T peak part.

4 ECG Signal Discussion

4.1 Our ECG Signals

4.2 Lead Comparison

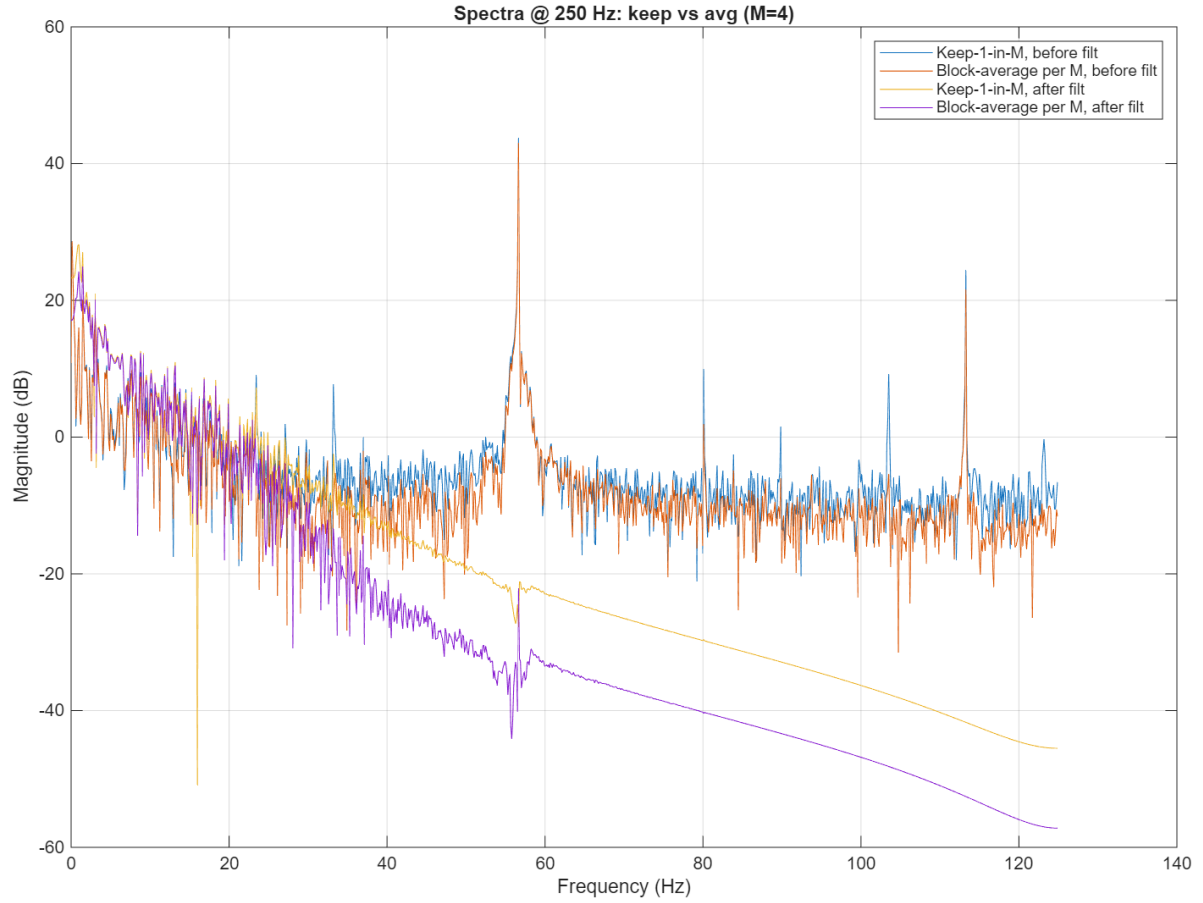


Figure 6: Frequency domain signal comparison of the two downsampling techniques, before and after filtering

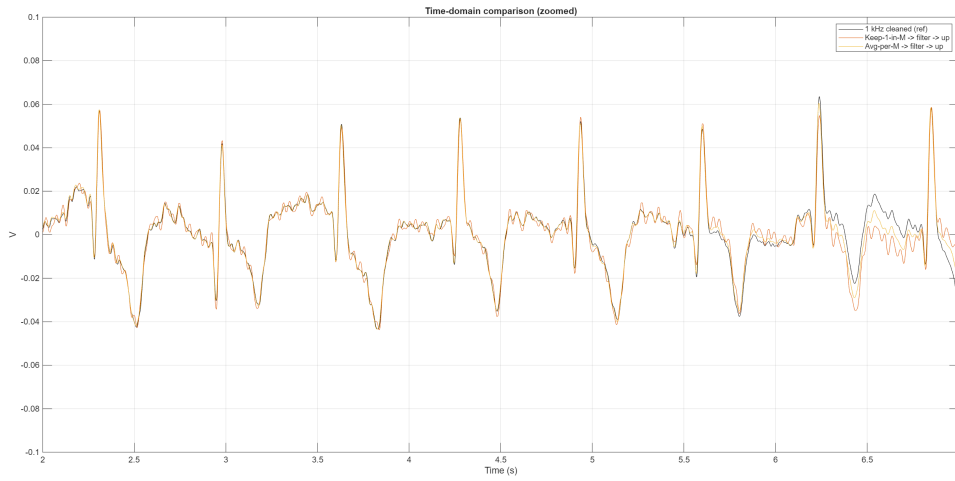


Figure 7: Time domain signal comparison of the three different signal processing pipelines (Oversampling, Direct downsampling and Averaging downsampling)

References

- [1] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Pearson, 2010.

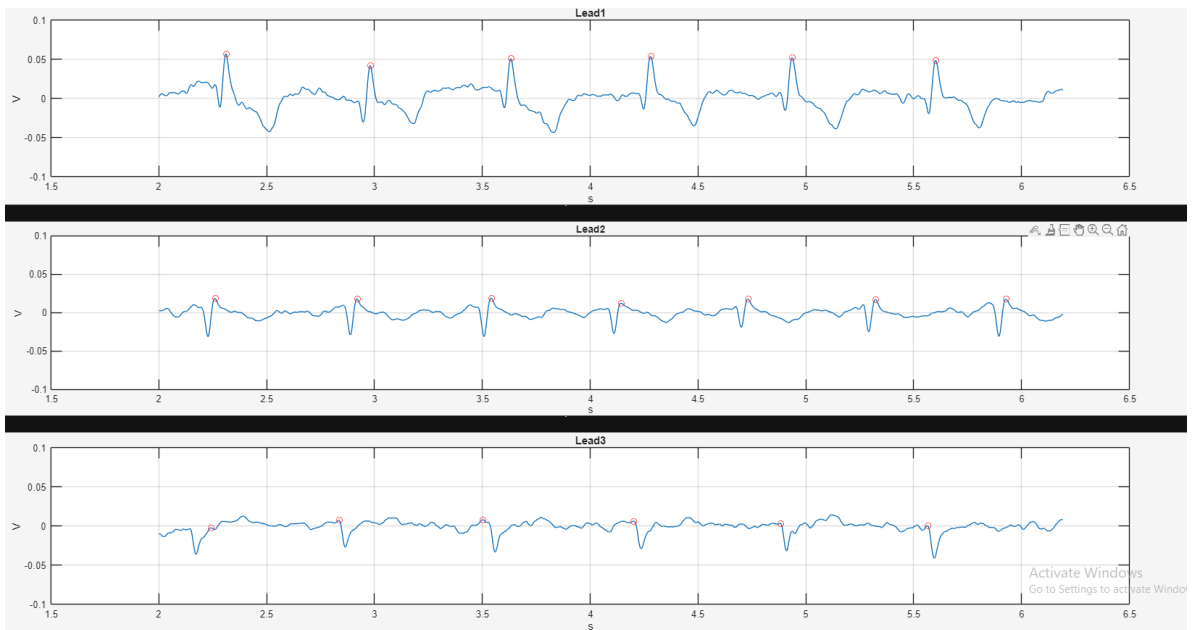


Figure 8: Comparing different Lead